

JACOB CARDINAL TREMBLAY

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EDUCATION

Max Planck Institute for Gravitational Physics (AEI) May 2024–May 2027
PhD in Natural Sciences, Leibniz Universität Hannover

Universität Bonn / Max Planck Institute for Radio Astronomy October 2021–April 2024
Master of Science in Astrophysics

West Virginia University August 2017–May 2021
Bachelor of Science in Physics

RESEARCH EXPERIENCE

Multi-Messenger Searches for Supermassive Black Hole Binaries May 2024–May 2027
Advisors: Dr. Rutger van Haasteren, Dr. Boris Goncharov, and Prof. Bruce Allen AEI

- PhD thesis at the Max Planck Institute for Gravitational Physics (AEI).
- Led the Parkes Pulsar Timing Array's targeted search for continuous gravitational waves from 3C 66B, a candidate supermassive black hole binary (SMBHB).
- Developed techniques to improve multi-messenger searches and introduced statistical quantities to understand the underlying tension between SMBHB models based on electromagnetic observations and PTA observations.
- Conducted more than 150 hours of pulsar observations with Murriyang, the Parkes 64-m radio telescope, for the Parkes Pulsar Timing Array.

Frequency Evolution Behaviour in Pulsars with the UBB Receiver March 2023–April 2024
Advisors: Prof. Dr. Michael Kramer and Dr. Ramesh Karuppusamy MPIfR

- MSc thesis at the Max Planck Institute for Radio Astronomy (MPIfR).
- Developed statistical methods to reliably identify and eliminate radio-frequency interference (RFI) in pulsar data from the Ultra-Broadband receiver (UBB).
- Leveraged the large frequency bandwidth of the UBB to study pulsar emission mechanisms by analyzing polarization and pulse-profile properties.

Optimal Channelization for Pulsar Dispersion Measurements January 2021–September 2021
Advisor: Prof. Maura McLaughlin West Virginia University

- Simulated observations of NANOGrav pulsars using the *Pulsar Signal Simulator* to create a model which takes into account different pulsar observational parameters and minimizes the uncertainty in dispersion measurements.

Analysis of Dispersion-Measure Variations in NANOGrav Pulsars May 2020–August 2020
Advisors: Dr. Cherry Ng and Dr. James McKee University of Toronto

- Manually removed RFI in high-cadence CHIME observations. Built Bash and Python pipelines and applied them to the clean CHIME data to construct two-dimensional pulse templates, and measure pulse times of arrival through template matching.
- Evaluated dispersion-measure variation statistics such as periodicity, Kolmogorov turbulence, solar-wind signatures, and evidence of discrete plasma structures in order to increase NANOGrav sensitivity.

PUBLICATIONS

1. **J. Cardinal Tremblay**, B. Goncharov, R. van Haasteren, et al. “Challenging Supermassive Black Hole Binary Candidates with Pulsar Timing Arrays.” *Submitted* (2026).
2. **J. Cardinal Tremblay** et al. “A Multi-Messenger Search for the Supermassive Black Hole Binary in 3C 66B with the Parkes Pulsar Timing Array.” *The Astrophysical Journal Letters*, **998**(2), L42 (2026).
3. S.-Y. Zhao, X. Zhu, **J. Cardinal Tremblay**, et al. “Targeted Search for Eccentric Supermassive Binary Black Holes in OJ 287 and Nearby Galaxy Clusters with PPTA DR3.” *Chinese Physics Letters*, **43**(5), 051102 (2026).
4. S. Sardana, B. Goncharov, and **J. Cardinal Tremblay**. “The Targeted Standard Siren Cosmology with Pulsar Timing Arrays.” *Preprint*, arXiv:2603.12168 (2026).
5. Y. Chen et al. (incl. **J. Cardinal Tremblay**). “Searching for Anisotropy in the Gravitational-Wave Background Using the Parkes Pulsar Timing Array.” *Physical Review D*, **113**(4), 043042 (2026).
6. M. Curyło et al. (incl. **J. Cardinal Tremblay**). “Frequency- and Phase-Resolved Polarimetry of Millisecond Pulsars and Its Application to Timing.” *Monthly Notices of the Royal Astronomical Society*, **549**(4), stag835 (2026).
7. S.-Y. Zhao, Z.-C. Chen, **J. Cardinal Tremblay**, et al. “Searching for Continuous Gravitational Waves in the Parkes Pulsar Timing Array Data Release 3.” *The Astrophysical Journal*, **992**(2), 181 (2025).
8. K. Aggarwal et al. (incl. **J. Cardinal Tremblay**). “Comprehensive Analysis of a Dense Sample of FRB 121102 Bursts.” *The Astrophysical Journal*, **922**(2), 115 (2021).

GRANTS AND AWARDS

BCGS Scholarship 2021–2023
Universität Bonn

- Scholarship awarded to the top applicants of the Bonn–Cologne Graduate School of Physics and Astronomy, covering living expenses during the MSc.

IAU Grant and Student KAS Grant 2022
International Astronomical Union / Korean Astronomical Society

- Received complementary IAU and KAS grants covering registration, accommodation, and additional costs for participation in the 2022 IAU General Assembly in Busan, South Korea.

BCGS Travel Grants 2022, 2023
University of Bonn

- Received BCGS travel grants for the 2022 IAU General Assembly in Busan and the 2023 Annual Meeting of the German Astronomical Society in Berlin.

SURP Fellowship Summer 2020
University of Toronto

- Research fellowship for the Summer Undergraduate Research Program (SURP) at the David A. Dunlap Department of Astronomy and Astrophysics and the Dunlap Institute for Astronomy and Astrophysics at the University of Toronto.

Ratna and Vaman Naik Endowed Physics Scholarship 2020–2021
West Virginia University

- Scholarship awarded to students who display excellence in academic merit.

National Space Grant Foundation Undergraduate Fellowship <i>West Virginia University</i>	2019–2021
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- NASA/National Space Grant Foundation Fellowship and scholarship to aid in funding undergraduate student research projects.

CSCAA Scholar All-American, First Team <i>West Virginia University</i>	2019–2020
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- Athlete achieving a GPA of 3.5 or higher and invited to the NCAA Swimming and Diving Championships.

Athletic Scholarship <i>West Virginia University</i>	2017–2021
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- Athletic scholarship to compete on West Virginia University’s NCAA Division I diving team.

International Student Scholarship <i>West Virginia University</i>	2017–2021
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- Scholarship awarded to international students with a GPA of 3.2 or higher.

COLLABORATION MEMBERSHIPS

Parkes Pulsar Timing Array (PPTA) <i>Member</i>	2024–Present
International Pulsar Timing Array (IPTA) <i>Member</i>	2024–Present
LISA Consortium <i>Community Member</i>	2026–Present

TEACHING AND MENTORING

Radio Astronomy Lab Course <i>Tutor</i>	April 2023–September 2024 <i>Universität Bonn</i>
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- Teaching assistant for the radio astronomy lab course. Guided MSc students in observing galaxies and pulsars with the Stockert 25-m Radio Telescope. Conducted oral examinations and assessed final laboratory reports.

Pulsar Search Collaboratory <i>Mentor</i>	September 2019–August 2021 <i>West Virginia University</i>
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- Tutored weekly classes during the Spring 2021 semester where I introduced high-school students to pulsar astronomy and helped them learn to search for pulsars.

ACADEMIC SERVICE AND OUTREACH

PhD Student Representative	April 2025–April 2026
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- Elected representative for the institute’s PhD students. Connected students with administrative, mental-health, and research support. Represented student interests in IMPRS meetings and to the Scientific Advisory Board. Helped organize lecture weeks, workshops, retreats, and social events.

Bonn Neutron Star Workshop <i>Local Organizing Committee Member</i>	May 2024 <i>Max Planck Institute for Radio Astronomy</i>
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- Participated in the planning, organization and logistics of the workshop.

MPIfR Sustainability Committee

Member

August 2023–May 2024

Max Planck Institute for Radio Astronomy

- Contributed to multiple sustainability efforts within the institute, including tracking the CO₂ footprint of research activities.

AIfA Social Committee

Member

September 2022–May 2024

Universität Bonn / AIfA

- Helped organize telescope nights, stargazing parties, as well as other social events for the members of the Argelander Institute for Astronomy (AIfA).

West Virginia University Astronomy Club

President

2020–2021

West Virginia University

- Gave biweekly interactive astronomy-related presentations, organized events such as telescope nights and stargazing parties, and helped guide younger students through their undergraduate careers.

Green Bank Observatory Open House

Department of Physics and Astronomy Ambassador

September 8, 2019

West Virginia University

- Gave physics demonstrations to children and the general public.

TECHNICAL SKILLS

Programming	Python, Bash, and L ^A T _E X
Observational	Murriyang, the Parkes 64-m radio telescope; AIfA 50-cm optical telescope; and WVU 36-cm optical telescope
Technical	Basic machining, soldering, and computer engineering
Languages	English and French (native); German (B1)

TALKS

Multi-Messenger Searches for Supermassive Black Hole Binaries <i>Conference Talk — European Astronomical Society Annual Meeting 2026</i>	<i>July 3, 2026</i> <i>Lausanne, Switzerland</i>
Multi-Messenger Searches for Supermassive Black Hole Binaries <i>Conference Talk — International Pulsar Timing Array Conference 2026</i>	<i>June 25, 2026</i> <i>Drakensberg, South Africa</i>
Multi-Messenger Searches for Supermassive Black Hole Binaries <i>Conference Talk — GRaviCon 2026: Gravitational Waves, the Age of Discovery</i>	<i>April 23, 2026</i> <i>Pisa, Italy</i>
Multi-Messenger Searches for Supermassive Black Hole Binaries <i>Conference Talk — Multi-Messenger Astrophysics in the Dynamic Universe</i>	<i>February 20, 2026</i> <i>Kyoto, Japan</i>
A Search for the Supermassive Black Hole Binary at the Center of 3C 66B in the Third Data Release of the Parkes Pulsar Timing Array <i>Colloquium — Max Planck Institute for Gravitational Physics</i>	<i>July 24, 2025</i> <i>Hannover, Germany</i>
A Multi-Messenger Search for the SMBHB in 3C 66B with the Parkes Pulsar Timing Array <i>Conference Talk — International Pulsar Timing Array Conference 2025</i>	<i>June 26, 2025</i> <i>Pasadena, USA</i>

Improving Pulsar Observations by Training a Machine-Learning Model on the EPTA Dataset <i>Poster — Bonn Neutron Star Workshop 2025</i>	<i>May 9, 2025</i> <i>Bonn, Germany</i>
Introducing Machine-Learning RFI Cleaning to Effelsberg Data Processing <i>Conference Talk — European Pulsar Timing Array Spring Meeting 2025</i>	<i>May 7, 2025</i> <i>Bonn, Germany</i>
Pulsar Timing Arrays <i>Invited Talk — AEI PhD Student Workshop</i>	<i>October 10, 2024</i> <i>Potsdam, Germany</i>
Frequency Evolution Behaviour in Pulsars with the Ultra-Broadband Receiver <i>Poster — International Astronomical Union General Assembly 2024</i>	<i>September 8, 2024</i> <i>Online</i>
Wideband Observations of Pulsar Radio Emission <i>Poster — Bonn-Cologne Graduate School of Physics and Astronomy Poster Session</i>	<i>April 19, 2024</i> <i>Bonn, Germany</i>
Wideband Observations of Pulsar Radio Emission <i>Conference Talk — Annual Meeting of the German Astronomical Society</i>	<i>September 12, 2023</i> <i>Berlin, Germany</i>
A Spectroscopic Multiplicity Survey of Galactic Wolf-Rayet Stars <i>Seminar Talk — University of Bonn</i>	<i>June 27, 2023</i> <i>Bonn, Germany</i>
Profile Evolution of B0355+54 with the Ultra-Broadband Receiver <i>Seminar Talk — Max Planck Institute for Radio Astronomy</i>	<i>March 21, 2023</i> <i>Bonn, Germany</i>
Optimal Frequency Channelization for Pulsar Dispersion Measurements <i>Poster — International Astronomical Union General Assembly 2022</i>	<i>August 5, 2022</i> <i>Busan, South Korea</i>
Dispersion-Measure Variations in NANOGrav Pulsars <i>Poster — Pulsar Search Collaboratory</i>	<i>July 30, 2021</i> <i>Morgantown, USA</i>
Optimal Frequency Channelization for Pulsar Dispersion Measurements <i>Talk — West Virginia Undergraduate Spring Symposium</i>	<i>April 17, 2021</i> <i>Online</i>
Dispersion-Measure Variations in NANOGrav Pulsars <i>Invited Talk — WVU Experimental Rocketry Club</i>	<i>February 16, 2021</i> <i>Online</i>
Dispersion-Measure Variations in NANOGrav Pulsars <i>Talk and Poster — University of Toronto Summer Undergraduate Research Program</i>	<i>Summer 2020</i> <i>Online</i>
An Analysis of the Sensitivity of the Arecibo Drift Scan Survey <i>Poster — West Virginia Undergraduate Spring Symposium</i>	<i>April 1, 2020</i> <i>Online</i>